

1.5 Accessibility Principles

رسائی پذیری (Accessibility) کے اصول یقینی بناتے ہیں کہ ہر انسان سسٹم استعمال کر سکے، بشمول مختلف جسمانی، حسی...

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 08 of 15

1.5 Accessibility Principles

LEARN IT

Accessibility principles are HCI guidelines that ensure digital systems are usable by **all** individuals, including people with diverse physical, sensory and cognitive abilities. They remove barriers to create equal access.

Accessibility is not a "special extra". It is how you make sure a student with low vision, a shopkeeper who cannot use a mouse, or a colour-blind driver can still complete the task.

Principle	What it means	Example
1. High contrast (واضح رنگ)	Make text and pictures stand out clearly against the background.	Black text on a white background in a railway app, readable in bright sunlight.
2. Readable fonts (پڑھنے میں آسان تحریر)	Clear, simple fonts with proper size and spacing.	A school website using large Calibri so homework instructions are easy to read.
3. Colour-blind friendly design (رنگوں کی پہچان)	Do not depend <i>only</i> on colours; also use symbols or text labels.	A traffic app showing a red circle with a "STOP" label, not just a red colour.
4. Inclusive navigation (آسان نیویگیشن)	Give users different ways to move around a system.	A website that lets you use the Tab key to jump between buttons if you cannot use a mouse.
5. Multimodal detection (متعدد حسی ذرائع)	Present information in different formats (sight, sound, touch).	A video with text captions, or a phone vibrating to alert a user.

WORKED EXAMPLE

Worked example: audit a school portal against the five principles.

1. **Contrast:** are result PDFs pale grey on white? Fail — use dark text.
2. **Fonts:** is the fee-challan note in 9-point decorative script? Fail — large Calibri (or a similar simple font), as the source recommends.
3. **Colour:** is "fail" only a red row with no word "Fail"? Fail — add the word, for colour-blind students.
4. **Navigation:** can a student who cannot use a mouse reach "Download date sheet" with Tab + Enter? If not, fail.
5. **Multimodal:** does the principal's video assembly have captions, and does a fee-due alert vibrate as well as flash? If not, deaf or busy students miss it.

The same five checks apply to an outdoor ATM (sunlight contrast) and to JazzCash (large numerals, Urdu labels, vibration on success).

REMEMBER IT

C-F-C-N-M: Contrast, Fonts, Colour-blind, Navigation, Multimodal. "Can Friends Catch New Messages?"

COMMON MISTAKE

Mistake: "Accessibility is only for people with disabilities, so our class portal does not need it."

Fix: The source says usable by **all** individuals. High contrast also helps in sunlight; captions help in a noisy bus; large fonts help tired eyes during board exams.

اردو میں سمجھیں

رسائی پذیری (Accessibility) کے اصول یقینی بناتے ہیں کہ ہر انسان سسٹم استعمال کر سکے، بشمول مختلف جسمانی، حسی اور ذہنی صلاحیتوں والے افراد۔ پانچ اصول: اونچا تضاد (کنٹراسٹ)، پڑھنے میں آسان فونٹ، صرف رنگ پر انحصار نہ کرنا، کی بورڈ سمیت متبادل نیویگیشن، اور معلومات کو نظر، آواز اور لمس کئی شکلوں میں دینا۔

CHECK YOURSELF

1. Why is a red-only "STOP" sign a problem?
2. Which principle does the Tab key support?
3. Give the source's school-website font example.

Answers: 1. Colour-blind users may not see the meaning if there is no symbol or "STOP" text.
2. Inclusive navigation. 3. Large Calibri on a school website for homework instructions.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Accessibility means:

- a. People with diverse abilities can use the system
- b. Only experts can use it
- c. Colourful GIFs always
- d. Mouse-only

MCQ 2. A principle is providing:

- a. Multiple ways to see/operate (text, caption, keyboard)
- b. Captcha of 12 steps
- c. 4pt grey on grey
- d. Auto-playing sound only

MCQ 3. Keyboard access helps:

- a. Motor impairment and power users
- b. Nobody
- c. Only AI models
- d. Only IoT soil

MCQ 4. Contrast helps:

- a. Low vision
- b. Faster bubble sort
- c. TCP window
- d. Primary keys

MCQ 5. Captions help:

- a. Deaf users and noisy labs
- b. Routers
- c. NAND
- d. K-maps

MCQ 6. Accessibility is part of:

- a. HCI quality, not an optional sticker
- b. Only marketing
- c. Only XII entrepreneurship
- d. Only physical cabling

Extra short questions

1. Define accessibility in HCI.
2. Four practical techniques.
3. Who else benefits besides disabled users?
4. Link to assistive tech in XI.

Extra long question

1. Audit a video lecture page against accessibility: list 6 fixes. Include one for vision, hearing, motor, language.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Design so people with disabilities (and many others) can perceive, operate and understand the UI.
2. Alt text, captions, keyboard, contrast, resizable text, Urdu/English toggle.
3. Broken arm, noisy room, bright sun, second-language users.
4. Screen readers need labelled buttons — that is an HCI job.

Long

1. Transcript+captions; contrast; pause; keyboard; alt on diagrams; Urdu summary. Do not autoplay. Test with Tab and a screen reader.

Next lecture: 1.6 Need Analysis for Interface Design